



# Open Automated Driving

Stack | Sim | Suite | Safety

IEEE ITSC 2026 | TUTORIAL



thinking cars

ika

RWTH AACHEN  
UNIVERSITY



|               |                                                                                                                                                     |
|---------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
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| 11:35 – 11:45 | <b>Why Collaborative OpenADS Development?</b><br>Timo Woopen, Thinking Cars GmbH                                                                    |
| 11:45 – 12:20 | <b>OpenADS Reference Architecture</b><br>Raphael van Kempen, RWTH Aachen University                                                                 |
| 12:20 – 13:00 | <b>OpenADS in Practice: Ecosystem Applications</b><br>Data Foundations   Benchmark-driven Qualification of OSS-Modules   OSS-based Fleet Management |
| 13:00 – 14:00 | <b>Lunch Break</b>                                                                                                                                  |
| 14:00 – 14:50 | <b>OpenADS Hands-On</b><br>Christian Geller, RWTH Aachen University                                                                                 |
| 14:50 – 15:00 | <b>Wrap-Up and Summary</b><br>Lessons learned and next collaboration steps.   All organizers                                                        |

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## Why Collaborative OpenADS Development?

Dr.-Ing. Timo Wooten

thinking cars  
ika | RWTH AACHEN  
UNIVERSITY

# Status Quo

## Automated Mobility is Reality – in the USA und China

### USA

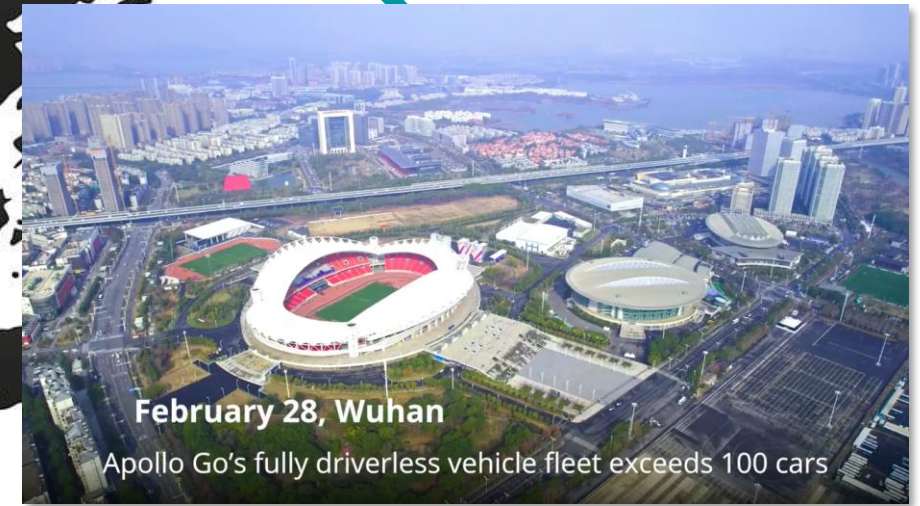
Driverless Robotaxi-Services in commercial operation



Source: <https://www.youtube.com/watch?v=8ebjO1HWQ9Q>

### China

Beijings demonstration area alone is 600 km<sup>2</sup>, ~900 vehicles and over 32 Mio. test-km.



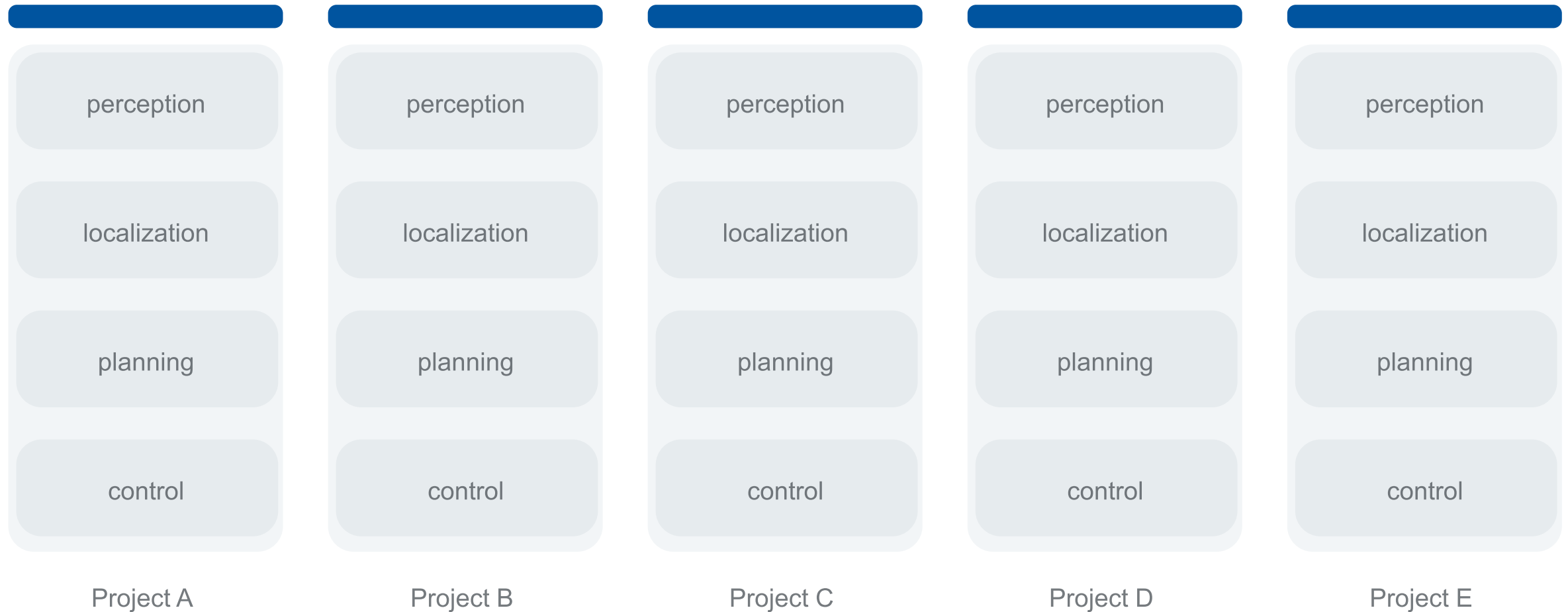
Source: <https://www.youtube.com/watch?v=lrZV4XVaSi4>

### Europe

No driverless operation with european AD-tech!

Europe's Challenge is not a capability gap; it's a coordination gap.

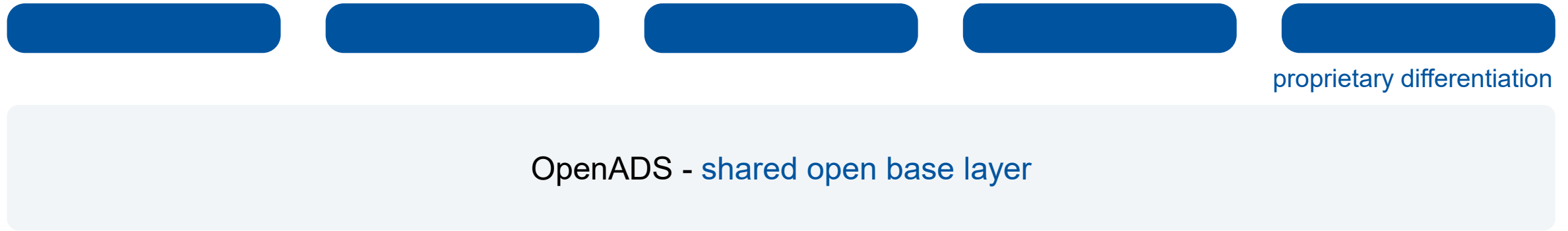
Research output, the suppliers and the safety expertise exist. It lacks is a shared foundation!



Each project rebuilds the same non-differentiating ~80% of the stack.

The thin bar is the differentiating remainder.

Europe's Challenge is not a capability gap; it's a coordination gap.  
Share the non-differentiating base, differentiate above it. Under European Sovereignty!



#### Notable Examples:

##### CLOUD INFRASTRUCTURE

Linux and Kubernetes became the shared substrate competitors build products on.

##### ROBOTICS

ROS and ROS 2 gave the field a common interface and message model.

##### SOFTWARE-DEFINED VEHICLE

Eclipse S-CORE and Eclipse SDV are organising a safety-relevant open base layer in automotive.

# European Initiative ECAVA – European Connected and Autonomous Vehicle Alliance



Commission boosts European automotive industry's global competitiveness

Quelle: [europa.eu](https://europa.eu)

## ECAVA'S FOCUS IN THE 4 WORKING GROUPS



### Software-Defined Vehicles:

Common AI-enabled software tools and open-source building blocks for digital vehicles.



### Automotive Hardware:

High-performance computing platforms, AI accelerators, chiplets and open-source standards



### AI and Data:

Data pooling, foundational AI models and a European autonomous driving stack.



### Autonomous Vehicle Deployment:

Cross-border piloting and real-world deployment on European roads.

### Key deliverables:

The working groups deliver yearly technology, implementation and business roadmaps, endorsed by the Steering Committee, with recommendations that champion industry-driven innovation and foster alignment with the EU and Member States on investment needs and regulation.



Source: [europa.eu](https://europa.eu)



## European Connected and Autonomous Vehicle Alliance (ECAVA)

Terms of Reference

August 2025



# German National Initiative Ecosystem Mobility 4.0

## 1 Community

building sustainable networks and cooperation structures with a genuine sense of belonging

## 2 Architecture

creating open integration and reference architectures

## 3 Technology

establishing open technology and methodology building blocks, including open-source solutions

## 4 Regulation and economy

aligning regulatory and economic frameworks and stimulating a growing provider market

## 5 Transfer

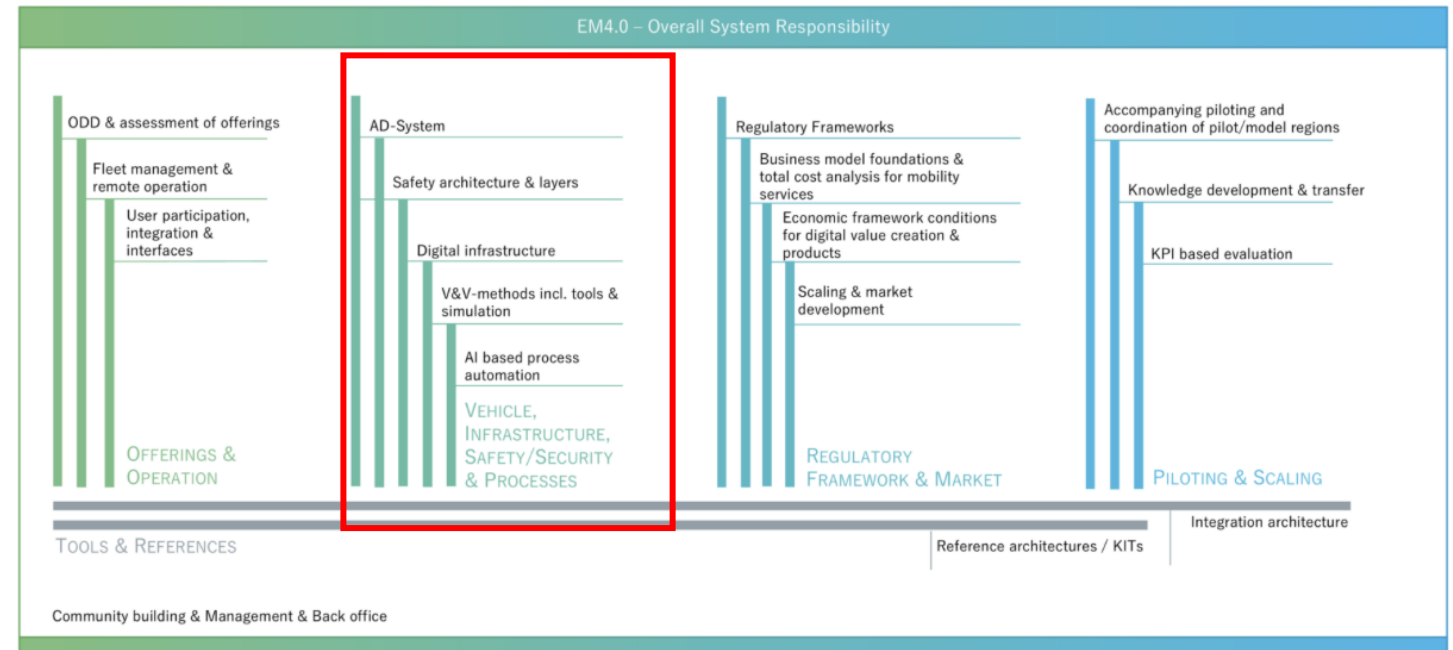
systematically integrating pilot projects, building knowledge and enabling scalable transfer, both nationally and internationally



Source: [Ecosystem Mobility 4.0](#)



## Fields of Action



Source: LinkedIn

## Ecosystem Mobility 4.0 and new industry players supply the missing layers.

### 01 | GOVERNANCE

Code without governance becomes abandonware.

?

### 02 | COMPOSABILITY

Code without a shared architecture does not compose.

?

### 03 | EVIDENCE

Code without reproducible evaluation cannot be integrated by anyone carrying safety responsibility.

?

# Ecosystem Mobility 4.0 and new industry players supply the missing layers.

## 01 | GOVERNANCE

Community building and back-office structures



**ECAVA**

## 02 | COMPOSABILITY

Reference architectures and an integration architecture. Vendor neutrality and data sovereignty as design principles



## 03 | EVIDENCE & INTEGRATION

e.g. Benchmark-driven Qualification of OSS-Modules

Aligned with the European Connected and Autonomous Vehicle Alliance.

# Ecosystem Mobility 4.0 and new industry players supply the missing layers.



**Open Automated Driving**  
Stack | Sim | Suite | Safety

## 02 | COMPOSABILITY

Reference architectures and an integration architecture. Vendor neutrality and data sovereignty as design principles

**OpenADS** first concrete technical output of the AD-System group

### OpenADStack

Modular ROS 2 reference implementation

### OpenADServices

Reusable modules: localization, perception, understanding, planning, control

### OpenADSim

Closed-loop simulation with CARLA and SUMO

### OpenADSuite

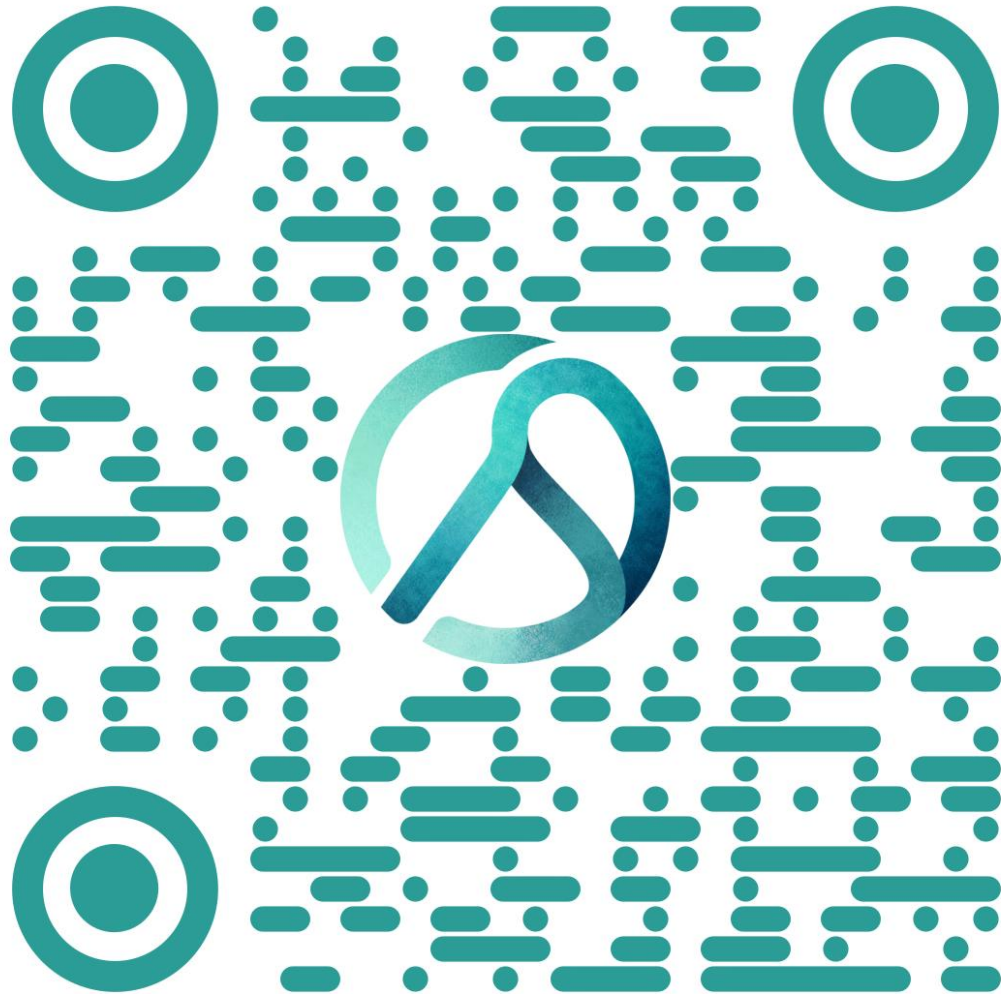
Development tooling

### OpenADSafety

Benchmarks and V&V tools

Aligned with the European Connected and Autonomous Vehicle Alliance.

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Information on the architecture, design principles and usage are summarized here:

<https://openads-project.github.io/>

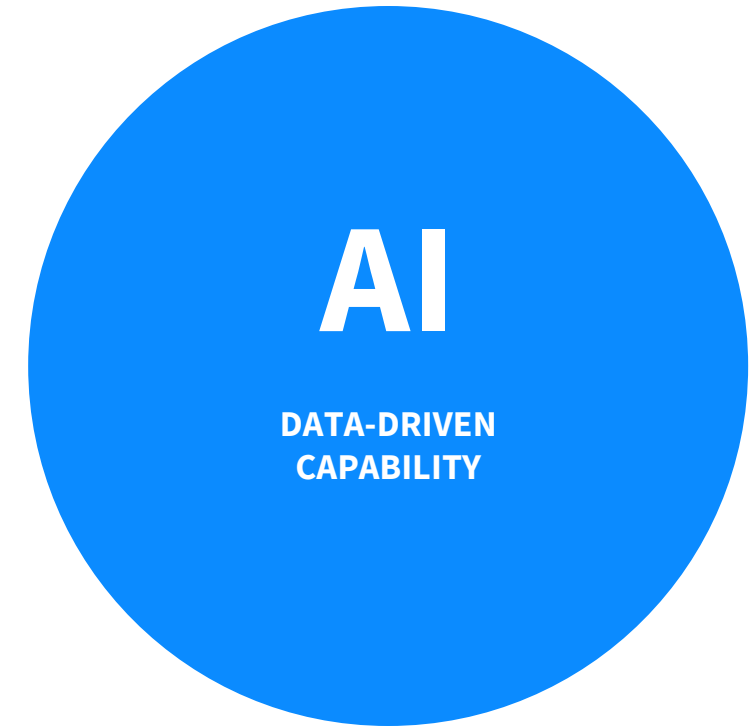
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# **FZI-AURA - Data as the Foundation for AI-Enabled Driving Functions**

# Open Source Is More Than Code

An AI driving function becomes reproducible only when its full technical context is accessible.

|                 |                                              |
|-----------------|----------------------------------------------|
| <b>CODE</b>     | reproducible implementation                  |
| <b>HARDWARE</b> | sensor and compute interfaces                |
| <b>DATA</b>     | real-world evidence for learning and testing |
| <b>MODELS</b>   | architectures and task definitions           |
| <b>WEIGHTS</b>  | learned capability                           |



Data is not an accessory to open AI. It is the evidence base that makes learning, comparison and validation possible.

# Transition in Autonomous Driving

From rule based to data driven

## Functions



# Different Data Layers



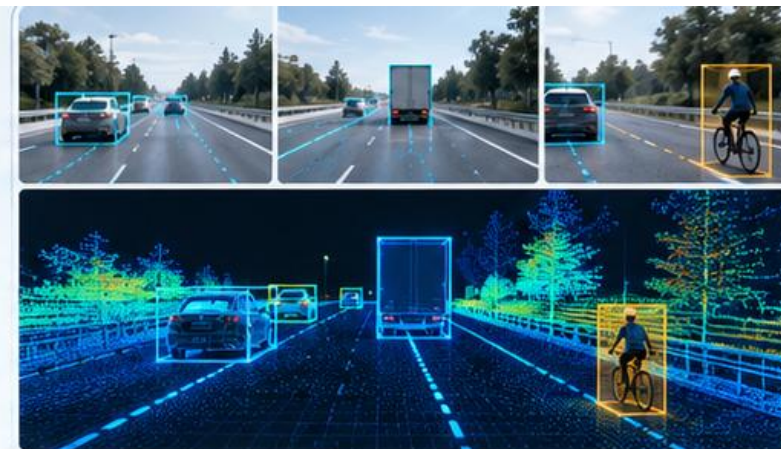
## ROUTING & LOCALIZATION

OSM and Lanelet2 maps provide topology, lane geometry and traffic rules.



## SCENARIO-BASED VALIDATION

scenario.center and OpenSCENARIO describe actors, maneuvers, triggers and conditions.



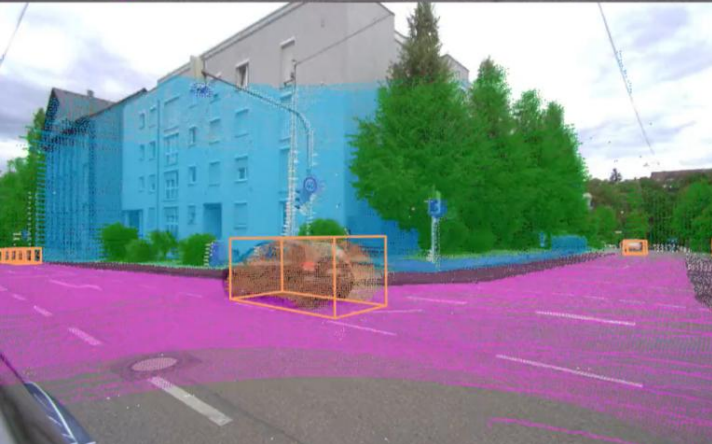
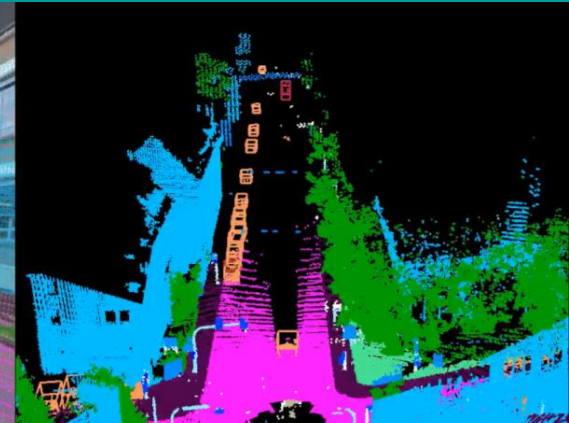
## AI PERCEPTION & DRIVING

Annotated camera, LiDAR, radar, ego and context data enable training and evaluation.



# FZI-AURA: Unprecedented LiDAR

The most labeled LiDAR points of any public non-synthetic autonomous-driving dataset!



**30.01B**  
semantically annotated  
LiDAR points

**4.11M**  
3D box annotations

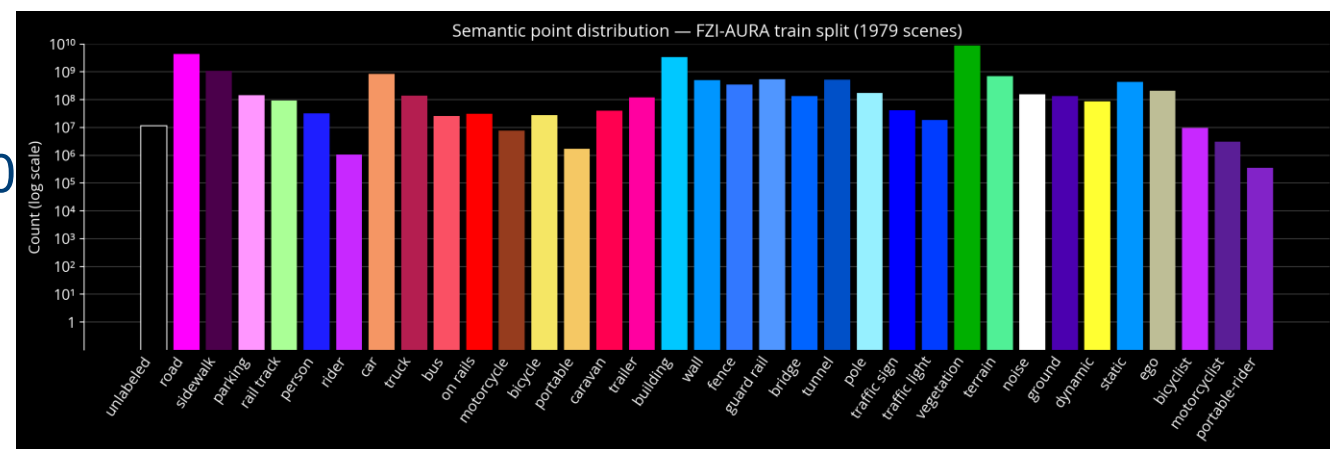
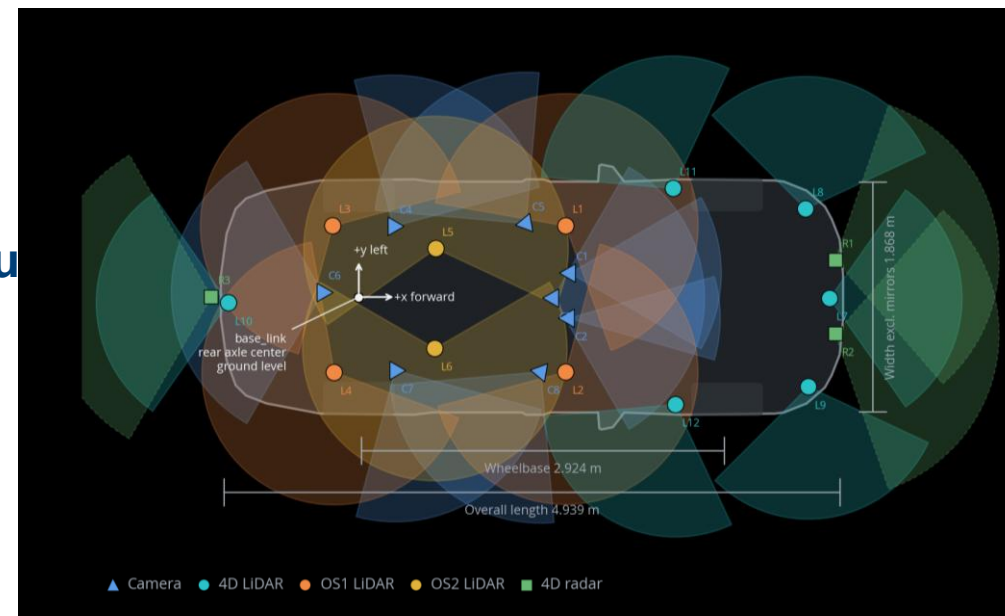
**2,473**  
scenes

**13.70 h**  
driving data

**493,754**  
samples at 10 Hz

# Multimodal Sensing Captures Real-World Variability

- 12 LiDARs
  - 2x OS2-128, 4x OS1-64, 6x Aeries II FMCW
  - **The largest public autonomous-driving LiDAR su**
- 8 Cameras
- 3 Radars
- Ego & Context
  - IMU, GNSS & ego-pose,
  - vehicle signals,
  - road and weather metadata
- Labeled key frames every 2 Hz, data in 10
  - Human LiDAR Semantic segmentation
  - Human 3D bounding boxes



# Open for Research and Commercial Use

## CC BY-SA 4.0

**USE SHARE ADAPT**

Data must remain usable after the funded project ends. Commercially reusable licensing prevents research results from becoming a dead end for industrial development.

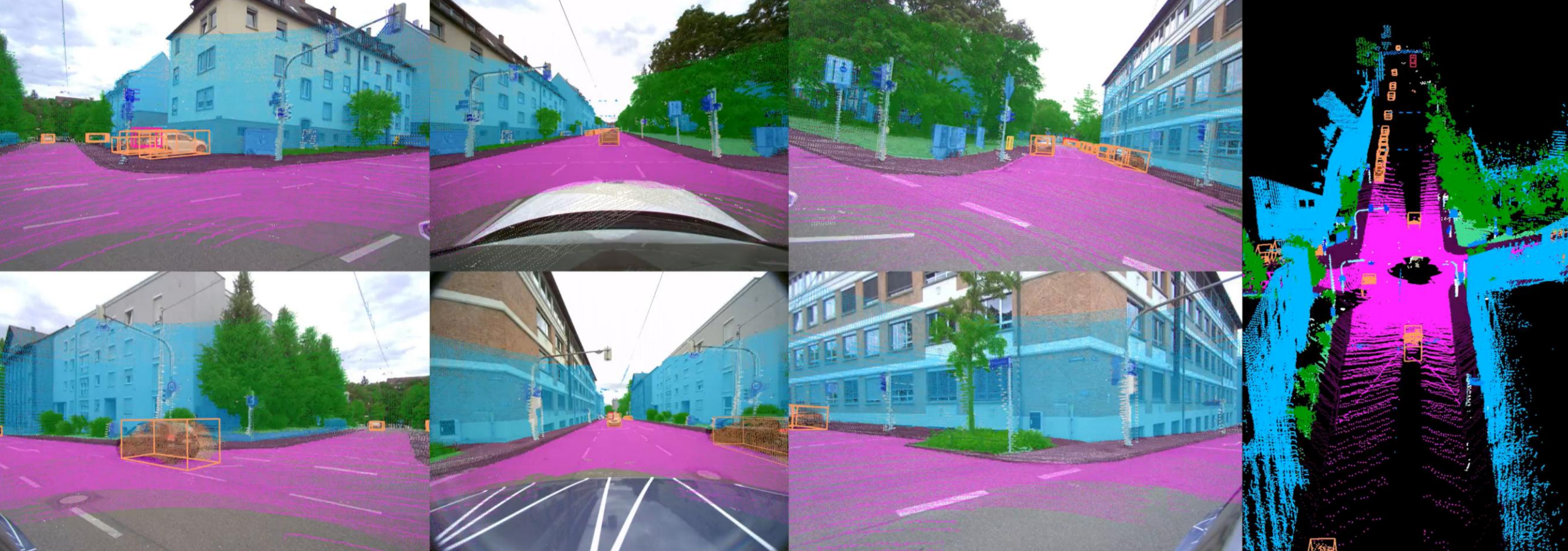
### Python SDK

Selective downloads, lazy scene and frame access, plus loading and projection helpers, compatible with PyTorch.



### Built in just better DATA (jbDATA)

FZI-AURA was created in the jbDATA project, funded by the German Federal Ministry for Economic Affairs and Energy and the European Union through NextGenerationEU.



**Get the dataset on Huggingface**

<https://url.fzi.de/FZI-AURA>

**Get in contact:**

Philip Schörner   Rupert Polley   Stefan Orf  
[schoerner@fzi.de](mailto:schoerner@fzi.de)   [polley@fzi.de](mailto:polley@fzi.de)   [orf@fzi.de](mailto:orf@fzi.de)



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# Orchestrating Autonomy

Benchmark-driven Qualification of  
OSS-Modules



# Why Establish Open-Source Software in Industry?

## Ecosystem Advantages on the Big Scale



**Speed** up development with code-first approach.



**Resilient** development of OSS excluded from restrictive regulations.

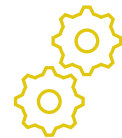


**Quality assured** by end-to-end control of large community.  
e.g. Linux

## Individual Economic Advantages

**OEM, TIER I and SW-Suppliers** can make use of scaling effects by reusing non distinguishing software and focus on relevant tasks.

**Service Providers** are enabled to offer services to a larger group of customers.



# Technological sovereignty is control over the AD roadmap.

If the AD stack sits inside an externally controlled platform, feature quality, feature timing, cost logic, and supplier leverage are often unclear.

**Roadmap control**  
Decide feature priorities and release timing on your own program logic.

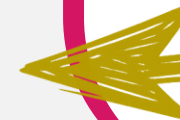
**Margin leverage**  
Reduce dependency-driven pricing power and protect long-term economics.

**Supplier optionality**  
Choose modules and partners by fit instead of inherited lock-in.

## CURRENT DEPENDENCY PATTERN

External control and dependency today

- Vehicle roadmap  
Adaptation speed follows platform boundaries and external priorities.
- Data and diagnostics  
Learning loops and debugging depth depend on supplier interfaces.
- Safety evidence  
Qualification logic is tied to assumptions you do not fully control.
- Commercial terms  
Scope expansion and total cost stay anchored in dependency.



## A concrete decision:

# Which object detection module do you build on?

No full stack in-house. The vehicle has to be composed from open modules, and each candidate only works in its own demonstration.

### Autaware Lidar CenterPoint

Center-based 3D detection, deployed in the Autaware perception pipeline

works in autaware only

### FocalFormer3D

Transformer detector with multi-stage focal queries, tuned for hard and missed objects

works in its own demo

### PillarOD

Pillar-based one-stage detector on a bird's-eye-view grid

works in its own demo



### ONE INTEGRATOR

Picks one, ships it, and carries the consequences for the service life of the vehicle.

Today the honest answer to “which one” is that nobody really knows.

## Status Quo of Module Selection:

Modules are selected by reputation, not by evidence.

Autoware Lidar CenterPoint

FocalFormer3D

PillarOD



ONE INTEGRATOR

- Demo videos on the maintainer's own recordings
- Benchmark numbers reported in papers, on incomparable setups
- Personal networks and conference impressions

### SAME METRIC NAME, TWO PAPERS

0.12 m

ATE RMSE · MODULE X

Dataset: highway sequences ODD: dry, daylight  
Tuning: per-sequence

0.11 m

ATE RMSE · MODULE Y

Dataset: campus loop ODD: mixed weather  
Tuning: default parameters



Comparable benchmarks cannot be assembled from published results. They must be designed and executed on one unified platform: [the same scenarios, the same metrics, the same ODD for every module.](#)

Illustrative. The numbers are close; they measure different things.

## Benchmark-driven qualification

Reproducible, scenario- & metric-based evidence, generated automatically per module

### PROPERTY 1

#### Comparability

Across contributions and across versions of the same module.

### PROPERTY 2

#### The module is the unit

Qualification applies to the module in a defined deployment, not to the vehicle.

A test tells you it ran. A metric is hard to compare.  
Our **benchmark** tells you how it compares, under stated conditions.

# Jointly build a sovereign open-source AD stack.

We propose an open marketplace: define, integrate, and qualify the stack you want based on open-source software and generate sovereignty in your software!

## DISCOVER

We identify the right open-source components for your target application and compliance requirements and map them against your existing stack.

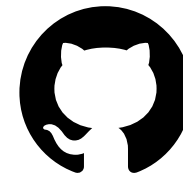
## EVALUATE & QUALIFY

We run reproducible benchmarks across real operating scenarios and safety conditions. Testing performance, safety, and reliability on your data for your auditable evidence.

## DEPLOY

We integrate open-source AD-software into your delivery pipelines. We support testing and homologation with our experienced partners.

Ecosystem already growing in OpenADS  
<https://github.com/openads-project/>



## What changes, by role.

|                           |                                                            |
|---------------------------|------------------------------------------------------------|
| Research                  | Comparable, citable module results.                        |
| OEM                       | Sovereign make-or-buy decided on measured evidence.        |
| Supplier                  | An entry ticket without prior OEM qualification.           |
| Integrator                | Evidence-based selection with quantified integration risk. |
| Assessor and homologation | Traceable evidence feeding ISO 26262 and SOTIF arguments.  |

# What changes, by role.

## Research

Comparable, citable module results.

---

Three ways to contribute.

### A module

Bring an implementation and let CI qualify it.

Support the industrialization and profit in licenses as it gets used in industry!

# What changes, by role.

## Research

Comparable, citable module results.

Three ways to contribute.

### A module

Bring an implementation and let CI qualify it.

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### A scenario

Add the case your ODD needs and others lack

# What changes, by role.

## Research

Comparable, citable module results.

Three ways to contribute.

### A module

Bring an implementation and let CI qualify it.

Support the industrialization and profit in licenses as it gets used in industry!

### A scenario

Add the case your ODD needs and others lack

### A metric

Define what your discipline considers evidence

|               |                                                                                                                                                                     |
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# Fleet Management APIs

AD STACK FROM A PRODUCT PERSPECTIVE

**From vehicle autonomy to fleet operations**

ITSC 2026 · OpenADS Tutorial

Markus Zarbock, Starwit Technologies



# Disclaimer

## PERSONAL PERSPECTIVE

The views and opinions presented in this talk are **my personal views**.

They:

- do not represent an official position of **Starwit Technologies**
- are not necessarily shared by my colleagues or partners
- are intended to stimulate discussion and exchange

**This is a technical and operational perspective, not a company statement.**



# Agenda

- Problem Description
- Two and a half examples
- Extending OpenADS
- Remarks and Opinions



# What's the problem?

- Fleet operator don't really care, how an AD stack works
- Relevant questions:
  - **Where are my (!) vehicles?**
  - **Which is available for rides?**
  - **Which one can serve next customer best?**
  - **Which distance can they travel before charging/refueling?**
  - **When is the next maintenance intervall?**
  - **Damage? Litter?**
  - **How to do Charging/Billing?**
- All questions are AD stack agnostic
- Necessary is a business driven interface



# Examples

- Street cleaning vehicles
  - Vehicles drive from A to A
  - Ride to target area, cleaning, ride back to depot
  - Vehicle or Robot? Both?
- Last mile shuttle service
  - multiple vendors
  - Just one aspect: Book a ride package & mobile conference room
- Intra logistics with AGV (autonomous guided vehicles) - [VDA 5050](#)



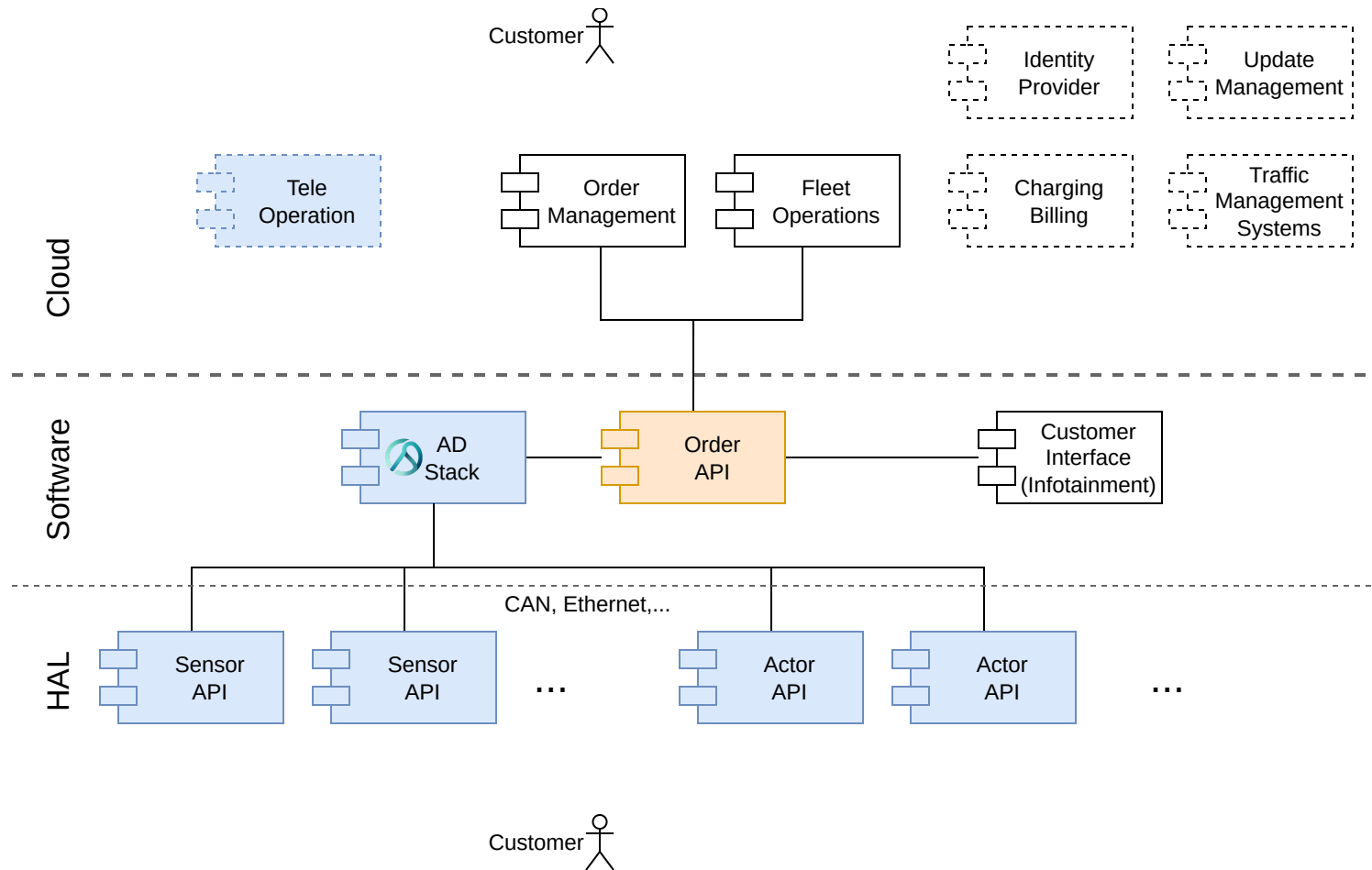


# How to enhance OpenADS

- Define first API version [OrderAPI](#)
- Implement an OpenADS module
- Test, fail, build next version
- **But:** Connectivity, Authentication & Authorization, Charging, ...
- Essential: OpenADS is available + simulation (!)



# Logical View





# API Example

```
RideOrder:
  type: object
  description: A ride order and its current execution state
  required: [orderId, source, stops, status, createdAt]
  properties:
    orderId:
      type: string
      format: uuid
    clientId:
      type: string
      description: Optional client-supplied idempotency key / external reference
      example: "fleet-job-88421"
    source:
      $ref: '#/components/schemas/OrderSource'
    stops:
      type: array
      description: >
        Ordered list of waypoints the vehicle shall visit. Must contain at
        least one stop (the final destination). The vehicle always departs
        from its current position at order acceptance time.
      minItems: 1
      items:
        $ref: '#/components/schemas/Stop'
```



# Food for thought

- Model vs API
- synchron vs asynchronous
- no safety, no real-time, not time critical
- [Loose Coupling](#)
- API versioning vs meta modeling (no!)
- Only agile will lead to results
- Order API will change more than everything else
- If you are interested: <https://github.com/starwit/Urbalytix>



# About Markus

<https://mastodon.social/@ztarbug> / <https://github.com/ztarbug>



**Markus**  
ztarbug

Edit profile

32 followers · 10 following

## Achievements



## Popular repositories

[Customize your pins](#)

### grpc-example

Public

An example showing how to connect C++, Rust and Python via gRPC and Protobuf

Python ☆ 4 🍴 1

### UserDataFilter

Public

Implements a Servlet filter to query LDAP/AD and put various user data on HTTP session.

Java ☆ 3 🍴 3

### object-detector

Public

Jupyter Notebook ☆ 3

### rust-examples

Public

Education repo with examples for the Rust programming language

Rust ☆ 3

### bath-experiments

Public

Python ☆ 2

### yolo-csharp

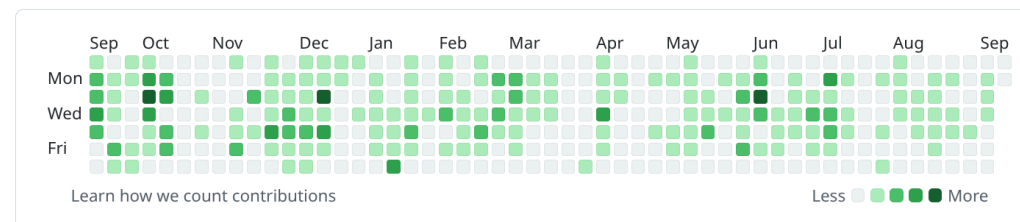
Public

Demonstrate how YoloDotNet can be used. Makes use of a pre-trained network in ONNX format.

C# ☆ 2

## 721 contributions in the last year

Contribution settings ▾



2026

2025

2024

2023

2022

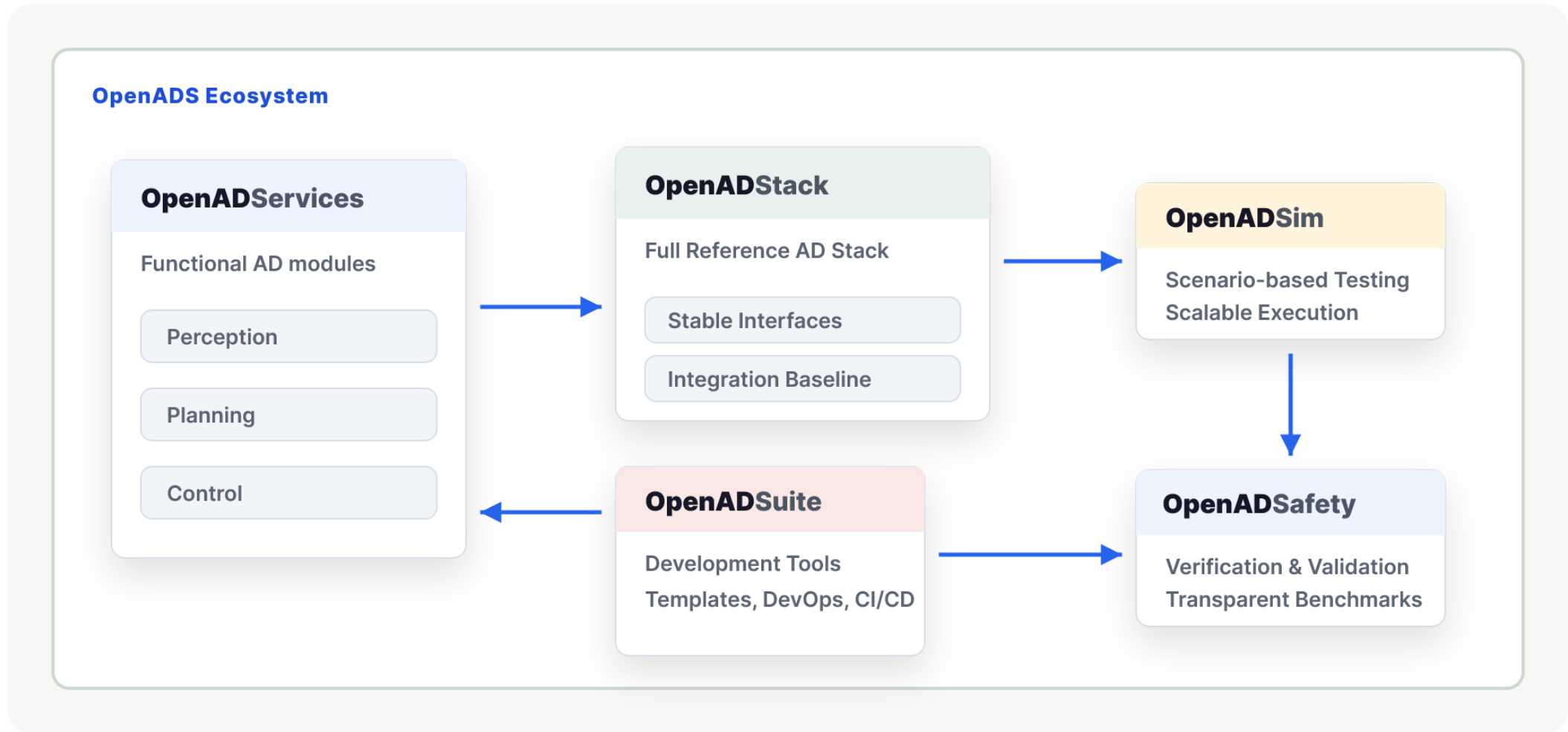


# Thanks for listening!

## QUESTIONS?

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| 12:20 – 13:00 | <b>OpenADS in Practice: Ecosystem Applications</b><br>Data Foundations   Benchmark-driven Qualification of OSS-Modules   OSS-based Fleet Management |
| 13:00 – 14:00 | <b>Lunch Break</b>                                                                                                                                  |
| 14:00 – 14:50 | <b>OpenADS Hands-On</b><br>Christian Geller, RWTH Aachen University                                                                                 |
| 14:50 – 15:00 | <b>Wrap-Up and Summary</b><br>Lessons learned and next collaboration steps.   All organizers                                                        |

# OpenADS Architecture



# OpenADS Hands-On



1

## Get Started in OpenADSim

Run and configure a deployment

2

## Investigate OpenADStack

See how services are connected

3

## Inspect an OpenADService

Review an exemplaric functional service

## OpenADSim

simulation environment based on  
CARLA or SUMO

## OpenADStack

baseline reference  
implementation

## OpenADService

federated modular building  
blocks

## Test with OpenADSim

Run scenario-based tests

6

## Integrate new Service

Add the service to a deployment

5

## Create your own Service

Use the shared development environment

4